Iverson Lai

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Education

Tufts University – *MS in Computer Science* — *Medford, MA* | graduated May 2026
University of California, San Diego – *BS in Environmental Science* — *La Jolla, CA* | graduated June 2022

Summary & Skills

Software engineer focused on C/C++, performance optimization, and ML infrastructure, with production full-stack experience.

- **Languages:** C, C++, Python, JavaScript/TypeScript
- **Systems/ML:** PyTorch, SIMD (AVX2, WASM SIMD), multithreading, INT8 quantization, WebAssembly, benchmarking
- **Web/Infrastructure:** Node.js, Express, Angular, MongoDB, AWS, nginx, REST APIs
- **Tools:** Git, Linux, CMake, GDB, Valgrind, GitHub Actions

Technical Projects

Machine Learning Library | *C, Python* — 2026

- Built a **neural network library from scratch in C** with no external dependencies beyond libc and libm, implementing custom matrix operations, reverse-mode automatic differentiation, and mini-batch SGD to train an MNIST classifier to **94% test accuracy in 14 seconds** on a laptop CPU.
- Implemented an autograd engine compiling computational graphs into flat forward and backward execution tapes via **iterative depth-first topological sorting**, with hand-derived analytical gradients for ReLU, softmax, and cross-entropy.
- Redesigned the core API to decouple graph definition from model compilation, replacing flag-driven side effects with an **atomic, validated constructor** preventing uninitialized or partial execution states.
- Hardened numerical stability and memory safety using **max-subtracted softmax**, clamped loss boundaries, strict tensor shape checks, and a versioned binary format; verified zero memory leaks or undefined behavior via **ASan and UBSan**.
- Managed memory via **custom arena allocators** built on virtual memory primitives (mmap/mprotect) with scratch pools for transient buffers, eliminating per-tensor allocations and fragmentation.

LLM Inference Engine | *C++17, PyTorch, WebAssembly* — 2026

- Built a C++ GPT-2 (124M) inference engine without BLAS or ML frameworks, validating numerical correctness against a hand-written PyTorch reference across **615 intermediate activations**.
- Accelerated decoding **373× to 42.6 tokens/s** on an Intel i5-1135G7 through KV caching, cache-blocked GEMM, AVX2 vectorization, and multithreading, reducing redundant computation and improving cache locality.
- Reduced model size from **498 to 129 MB** using per-channel INT8 quantization; isolated embedding outliers and retained 8 columns in FP32 to achieve **97.4% top-1 agreement** with the FP32 baseline.
- Delivered single-threaded browser inference at **21 ms/token** using **WebAssembly SIMD** and INT8 weights, with a JavaScript tokenizer validated against Hugging Face and deployment gates for numerical regression tests.

Re: Genesis | *Godot 4.4, GDScript* — 2025

- Developed a **utility-based tactical AI** that ranks movement, attack, healing, and buff actions using damage potential, target health, aggression, positional safety, and ally proximity, with fallback execution when preferred actions fail.
- Implemented **terrain-weighted grid searches** with movement-cost relaxation, obstacles, and faction-aware occupancy rules, combining reachable positions with weapon and ability ranges to generate tactical overlays.
- Coordinated campaign and battle state machines with an **asynchronous UI queue**, sequencing dialogue, camera movement, and notifications while blocking gameplay input until queued actions complete.

Professional Experience

Ming Long Insurance — Remote
Freelance Software Engineer — Sept 2024 – Present

- Took over engineering for a Medicare insurance agency's **production software**, including an Angular/Node.js/MongoDB agent portal and public website deployed on AWS behind nginx.
- Planned a **phased modernization** from Angular 14→22, Node 18→24, Express 4→5, and Mongoose 6→9, sequencing releases around Medicare open enrollment to minimize production risk.
- Restored a frontend build broken for 2.5 years after automated dependency updates mixed incompatible Angular versions, then **reduced known dependency vulnerabilities 28%** without altering production JavaScript.

Truvian Science — San Diego, CA
Clinical Development Intern — June 2024 – Sept 2024

- **Aggregated and validated** blood-test results from Truvian, Cobas, and Sysmex analyzers, investigating anomalous measurements from clinical development studies' before downstream study analysis and reporting.
- Performed **daily QC and calibration** across 5+ Truvian prototype devices, supporting repeated experimental runs and comparing device results against industry standard Cobas and Sysmex clinical analyzers.